PYTHON PROGRAMMING



21 Days Challenge

Day-20

Day-20

SENDING EMAILS WITH PYTHON
&
WEB SCRAPING USING PYTHON



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SENDING EMAILS WITH PYTHON

Import Library

Create Message

Setup SMTP Server

Introduction

- Sending Emails with Python
- Python can send emails using the SMTP protocol.
- Real-world use: notifications, automation, reports.

Import Library

```
import smtplib
```

- smtplib → Python's built-in library for sending emails.
- SMTP = Simple Mail Transfer Protocol.

Create Message & Format

```
s_email = input("Enter Sender Mail:")
r_email = input("Enter Receiver Mail:")
subject = input("Enter Subject:")
message = input("Enter Message:")


text = f"Subject:{subject}\n\n{message}"
```

SMTP Server Setup

```
server = smtplib.SMTP("smtp.gmail.com", 587)
server.starttls()
```

- smtp.gmail.com = Gmail's mail server.
- Port 587 = TLS security.
- starttls() → encrypts the connection.

Login to Server

```
server.login(s_email, "app-password")
```

- Use App Password, not your real Gmail password.
- Google Account → Security → App Passwords.

Send Email & Close Connection

```
server.sendmail(s_email, r_email, text)

server.quit()
```

- Need internet connection.
- Must enable App Passwords in Gmail.
- Receiver may check spam folder if not found.

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WEB SCRAPING USING PYTHON

Install & Import Libraries

Fetch a Web Page

Extract Specific Data

Introduction

- **requests** → to send HTTP requests and get page content.
- **BeautifulSoup** → to parse and navigate HTML.
- **lxml** → a fast HTML/XML parser used by BeautifulSoup.

Install & Import Libraries

```
!pip install requests beautifulsoup4 lxml
```

```
import requests
from bs4 import BeautifulSoup
```

Fetch a Web Page

```
url = "https://webscraper.io/test-sites"
response = requests.get(url)


print(response.status_code) # 200 means success
#print(response.text[:500])
```

Parse HTML with BeautifulSoup + lxml

```
soup = BeautifulSoup(response.text, "lxml")
# Get page title
print("Page Title:", soup.title.string)


# Get all links
links = soup.find_all("a")
for link in links[:5]: # print first 5 links
  print(link.get("href"))
```

Extract Specific Data (Example)

```
headings = soup.find_all("h2")
for h in headings:
  print(h.text.strip())
```

QUESTION SESSION

THANK YOU

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